UNIT V

①

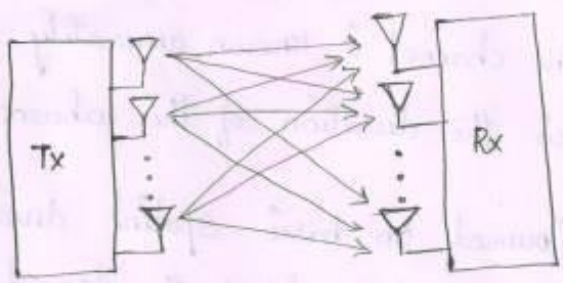
Multiple Antenna Techniques

MIMO systems :-

- Multiple input multiple output or MIMO is a radio communication technology or RF technology that is being mentioned and used in many new technologies these days.

- The wireless MIMO channel is assumed to consist of a system with multiple transmit antennas and multiple receive antennas which are connected by means of fading channels

MIMO Model



- The antennas are placed in order to ensure that the channels across antennas are independent.

- placement separation in units of $\lambda/2$ is one means to ensure independences. conventionally for a MIMO system with M transmit antennas & N receive antennas the number of channels counting each link from a transmitter to a receiver separately is MN much like in conventional Wireless Communication.

- These channels must be known at least the receiver or transmitter or both in order to communicate information successfully during the coherence interval.

- This is done by means of estimation of the channel coefficients using an appropriate techniques

- An M transmitter N receiver MIMO system represented as an NxM matrix denoted by H.

- A flat fading model for simplicity of analysis since transform techniques such as OFDM can be used to convert frequency selective channels into flat fading ones.

- Assume that the channel is known accurately to the receiver & does not change in the duration of this coherence interval.

- MIMO system focused on basic spatial diversity here the MIMO system was used to limit the degradation caused by multipath propagation.

- The first step as system then started to utilise the multipath propagation to advantage turning the additional signal paths into what might effectively be considered as additional channels to carry additional data.

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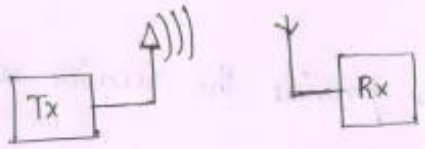
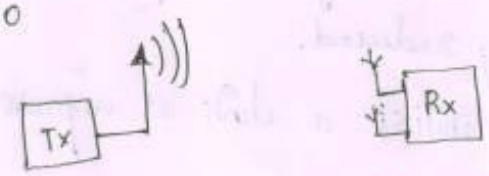
Multi antenna types

SISO:-



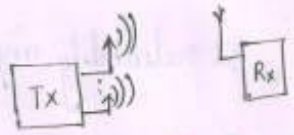
Single i/p single o/p means that the transmitter and receiver of the radio system have only one antenna.

SIMO



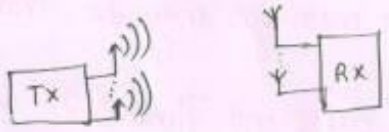
Single input multiple output means that the receiver has multiple antennas while the transmitter has one antenna.

MISO



Multiple input single output means that the transmitter has multiple antennas while the receiver has one antenna

MIMO



Multiple i/p multiple o/p means that the both the transmitter & receiver have multiple antennas.

- The channel may be affected by fading & this will impact the signal to noise ratio.
- The principle of diversity is to provide the receiver with multiple version of the same signal.
- If these can be made to be affected in different ways by the signal path, the probability that they will be affected at the same time is considerably reduced.
- The diversity helps to stabilise a link & improve proformance reducing error rate.
- Multiple data streams transmitted in a single channel at the same time
- Multiple radios collect multipath signals
- Delivers simultaneous speed, covererage, & reliability improvements

Several different diversity modes are available & provide a number of

Time diversity :-

Using time diversity a message may be transmitted at different times e.g. Using different time slots & channel coding

FREQUENCY Diversity:- (3)

This form of diversity uses different frequencies. It may be in the form of using different channels or technologies such as spread spectrum/OFDM

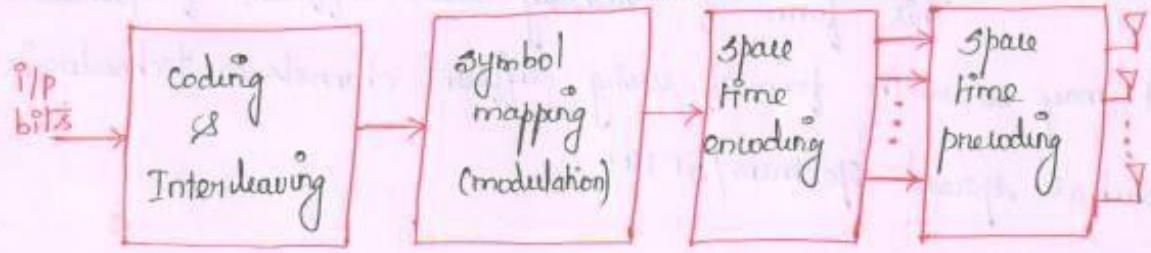
Space diversity:

Space diversity used in the broadest sense of the definition is used as the basis for MIMO. It uses antennas located in different positions to take advantage of the different radio paths that exist in a typical terrestrial environment.

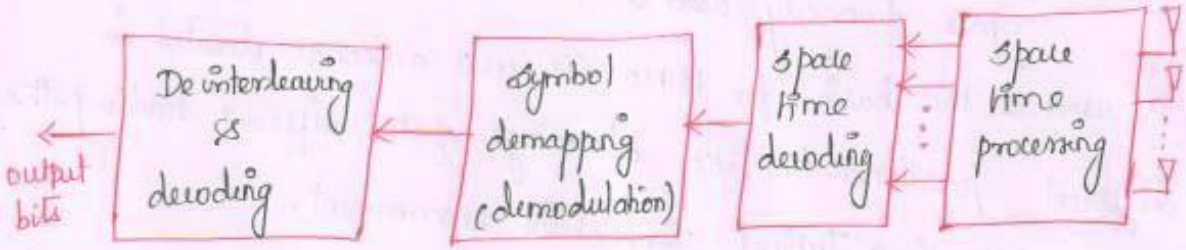
- MIMO uses multiple antennas on both the transmitter and receiver. They have dual capability of combining the SIMO & MISO technologies.
- They can also ↑ capacity by using spatial multiplexing.
- The MIMO method has some clear advantages over SISO methods
- The fading is greatly eliminated by spatial diversity low power is required compared to other techniques in MIMO.
- The number of antenna element ↑ the channel capacity is ↑
- The improving of MIMO from MISO channel capacity as antenna ↑.

Building Blocks of MIMO

Transmitter



Receiver



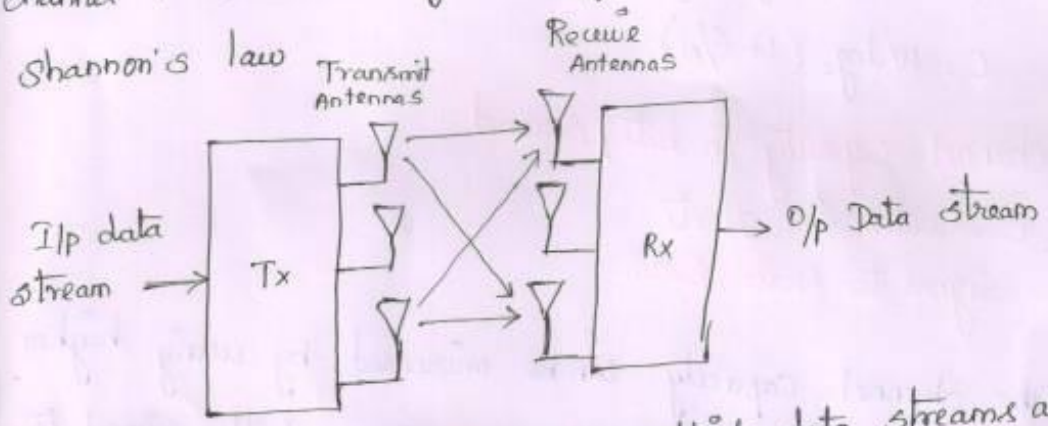
- The space dimensions to improve wireless systems capacity range & reliability
- It offers significant ↑ in data throughput & link range without additional bandwidth or increased transmit power.
- MIMO achieves this goal by spreading the same total transmit power over the antenna to achieve an array gain that improves the spectral efficiency or to achieve a diversity gain that improves the link reliability.

Spatial Multiplexing (4)

One of the key advantages of MIMO spatial multiplexing is the fact that it is able to provide additional data capacity.

- MIMO spatial multiplexing achieves this by utilising the multiple paths & effectively using them as additional channels to carry data.

- The maximum amount of data that can be carried by a radio channel is limited by the physical boundaries defined under Shannon's law



- The spatial multiplexing, multiple data streams are transmitted at the same time.

- They are transmitted on the same channel but by different antenna. They are recombined at the reciever using MIMO signal processing.

Shannon's law :-

- The amount of data that can be passed along a specific channel in the presence of noise.

- The law that governs that is called Shannon's law named after the formulated it.

Shannon's law defines the maximum rate at which error free data can be transmitted over a given bandwidth in the presence of noise. It is usually expressed in the form

$$C = W \log_2 (1 + S/N)$$

C - channel capacity in bits / second

W - Bandwidth in Hertz

S/N - Signal to Noise Ratio

- The channel capacity can be increased by using higher order modulation scheme but these require a better signal to noise ratio then the lower order modulation schemes. but these require a better signal to noise ratio then the lower order modulation schemes

- Thus a balance exists b/w the data rate & the allowable error rate signal to noise ratio & power that can be transmitted.

- Spatial Multiplexing is a transmission technique in MIMO spatial multiplexing is a wireless communication to transmit independent & separately encoded data signals so called streams for each of the multiple transmit antennas. (5)

- If the transmitter is equipped with N_t antennas & the receiver has N_R antennas the maximum spatial multiplexing order is $N_S = MIN(N_T, N_r)$

N_S – Streams can be transmitted in parallel, ideally leading to an N_S ↑ of the spectral efficiency.

- The multiplexing gain can be limited by spatial correlation which means some of the parallel stream may have very weak channel gains.

BEAM FORMING:-

- Antenna technologies are the key in increasing network capacity. It started with sectorized antennas.

- These antenna illuminate 60° or 120° operate as one cell.

- Adaptive antenna arrays intensify spatial multiplexing

- Smart antennas belong to adaptive antenna arrays but differ in their smart direction of arrival estimation
- Smart antennas can form a user specific beam. optional feed back can reduce complexity of the array system

Beam forming is the method used to create the radiation pattern of an antenna array. It can be applied in all antenna array systems as well as MIMO systems.

Smart antennas are divided into 2 groups.

* phased array system
* Adaptive array system with an infinite number of patterns adjusted to the scenario in real time

Switched Beam former

Adaptive Beam former

- Switched beam formers electrically calculate the DOAs switch on the fixed beam. The user only has the optimum signal strength along the center of the beam.
- The adaptive beam former deals with that problem's adjusts the beam in real time to the moving UE. The complexity & the cost of such a system is higher than

H. First type

Pre coding

(6)

- It is a generalization of beam forming to support multilayer transmission in multi antenna wireless communications.

- In conventional single layer beam forming the same signal is emitted from each of the transmit antennas with appropriate weighting such that the single power is maximized at the receiver o/p.

Pre coding can be separated by two classifications

* pre coding for single user MIMO
* precoding for Multi user MIMO

Pre coding for single user MIMO

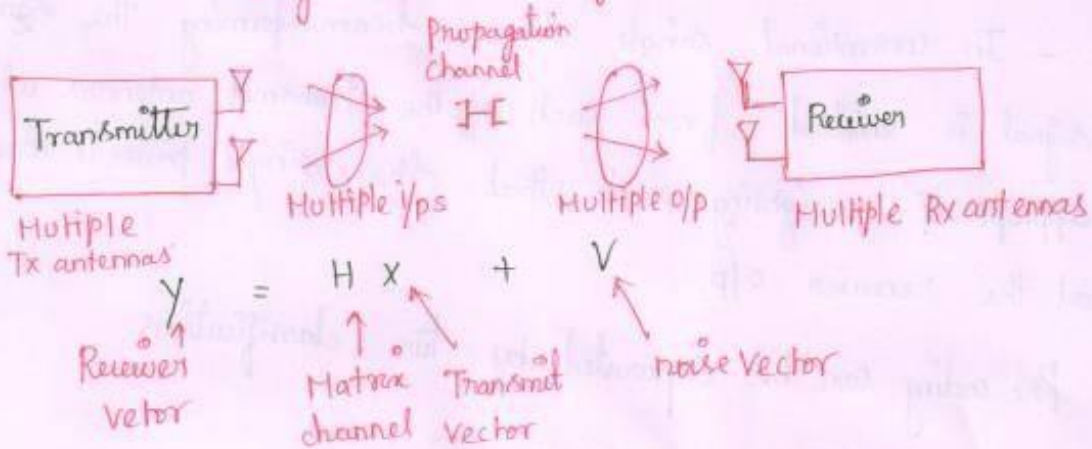
- In single user MIMO systems a transmitter equipped with multiple antennas communicate with a receiver that has multiple antennas.

- Most classic precoding assume narrowband, slowly fading channels meaning that the channel for a certain period of time can be described by a single channel matrix which does not change faster.

- The precoding strategy that maximize the through put called channel capacity depends on the channel state information available in

- Single user MIMO communication systems exploit multiple transmit and receive antennas to improve capacity, reliability & resistance to interference.

Single User MIMO System



Pre coding for Multi User MIMO

- In multi User MIMO a multi antenna transmitter Communicates Simultaneously with multiple receivers. This is known as Space division multiple access.

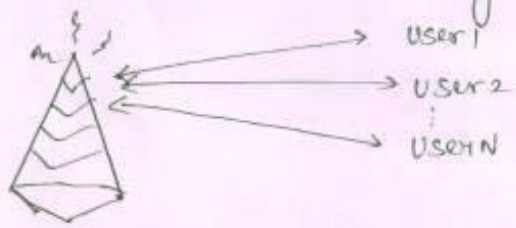
- precoding algorithm for SDMA System can be Sub-divided into linear & non linear pre coding types.

- The capacity achieving algorithms are non linear but linear pre coding approaches usually achieve resonable performance with much lower Complexity.

- Linear precoding strategies include MMSE precoding & the Simplified zero forcing precoding. (7)

- There are also precoding strategies tailored for low-rate feedback of channel state information for ex random beam forming.

- Non linear precoding is designed based on the cocept of dirty paper coding which shows that any known interference at the transmitter can be subtracted without the penalty of radio resources if the optimal pre coding scheme can be applied on the transmit signal.



Diversity coding :-

- It is used when there is no channel knowledge at the transmitter.

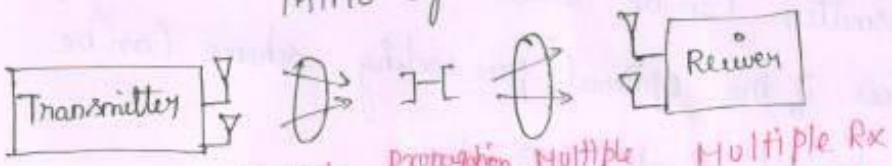
- In diversity methods a single stream is transmitted but the signal is coded using techniques called space time coding.

- The signal is emitted from each of the transmit antennas with full or near orthogonal coding.

CHANNEL MODEL (SYSTEM MODEL)

- The transmitter & receiver are equipped with multiple antenna elements.
- The transmit stream go through a matrix channel which consists of multiple receive antennas at the receiver.
- Then the receiver gets the received signal vector by the multiple receive antennas & decodes the received signal vectors into the Original information

MIMO System Model



$$r = H s + n$$

Noise vector

Receive vector — matrix channel — transmit vector

- r is the $M \times 1$ received signal vector as there are M antennas in receiver
- H represented channel Matrix
- s is the $N \times 1$ transmitted signal vector as there are N antennas in transmitter
- n is an $M \times 1$ vector of additive noise term

Let Q denote the covariance matrix of x then the capacity of the system described by information (8)

$$C = \log_2 [\det (I_M + HQH^*)] \text{ b/s/Hz}$$

- This is optimal when is known at the transmitter & the i/p distribution maximizing the mutual information is the Gaussian distribution

- channel feedback may be known at the transmitter & the optimal is not proportional to the identity matrix but is constructed from a water filling argument.

- The effect of $Q = (P/N)$. Q based on perfect channel estimation & feedback then we can evaluate a maximum capacity gain due to feedback.

- The effect of H matrix has wide range of channel models including for ex. correlated fading & specular components,

$$C_{EP} = \sum_{L=1}^{M} \log_2 \left(1 + \frac{P}{N} \lambda_i\right) \text{ b/s/Hz}$$

where $\lambda_1, \lambda_2 \ldots \lambda$ the non zero eigen values of W

$m = \min(M, N)$

$$W = \begin{cases} HH^*, & M \le N \\ H^*H, & N \le M \end{cases}$$

The singular Value Decomposition goes by

$$H = UDV^*$$

U & V are Unitary

$$D = \text{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \sqrt{\lambda_3}, \ldots \sqrt{\lambda_m}, 0, \ldots 0)$$

The MIMO signal model

$$\hat{r} = D\tilde{s} + \tilde{n}$$

where $\tilde{r} = U^* r$, $\tilde{s} = V^* s$ & $\tilde{n} = U^* n$

When the channel is known at the transmitter then H is known at above eqn

Capacity over Q subject to the power constraint $tr(Q) \leq P$

The optimal Q in this case is well known & is called a water filling solutions.

Capacity

$$C_{wp} = \sum_{i=1}^{M} \log_2(\mu\lambda_i)^+ \text{ b/s/Hz}$$

where μ is chosen to satisfy

$$P = \sum_{i=1}^{M} (\mu - \lambda_i^{-1})^+$$

MIMO Diversity Techniques (9)

- Diversity can be implemented at the transmit end at the receive end or at both ends of the wireless link.

- Generally MIMO diversity techniques can provide higher SNR and improve transmission reliability.

Transmit diversity :-

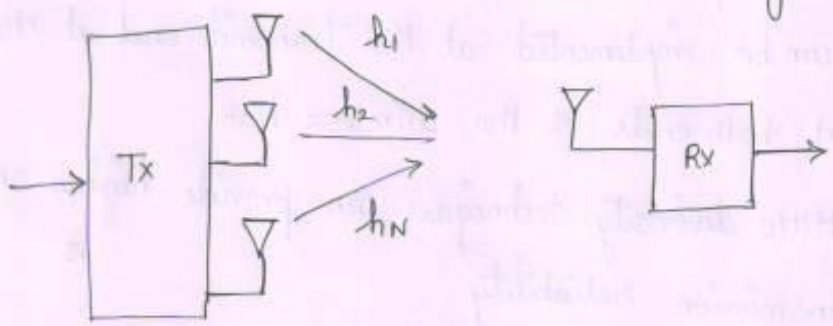
- It improves the signal quality and achieves a higher SNR ratio at the receiver side; it involves transmitting data stream through multiple antennas & receiving by single antenna or more.

- Transmit diversity can effectively mitigate multipath fading effects as multiple antennas afford a receiver several observations of the same data stream.

- Each antenna will experience a different interference environment & if one antenna experienced a deep fade, then it is likely that another has a sufficient signal.

- Ex: The transmit diversity techniques include Alamouti code & orthogonal j codes proposed by whole system Nt transmit antenna system.

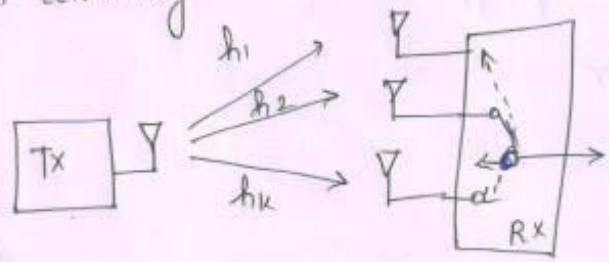
Transmit Diversity



Receive Diversity

- It is widely used in wireless communication systems it can be achieved by receiving redundant copies of the same signal.
- The idea behind receive diversity is that each antenna at the receive end can observe an independent copy of the same signal.
- The probability that all signals are in deep fade simultaneously is significantly reduced.
- This type of diversity hasn't particular settings or requirements on the transmit end but requires a receiver that could simultaneously process all received signals & combines them by a proper combining method.
- There are several classical methods for combining the different diversity branches at the receiver most important of which & most widely used are selection combining, Maximal Ratio combining & equal gain combining

Selection combining (10)



- The selection diversity improve SNR.

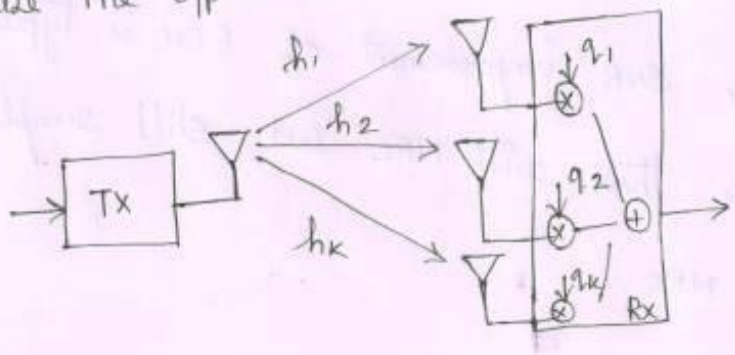
Let $\tilde{\gamma}$ is new average signal SNR. Γ_{avg}. SNR in each branch

$$\tilde{\gamma} = \gamma \sum_{K=1}^{M} \frac{1}{K} = \gamma \left(1 + \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} + \cdots \tfrac{1}{M}\right) > \Gamma$$

Maximal Ratio Combining

- The selection combining technique ignores information from all diversity branches except the particular branch that has highest SNR. This drawback is mitigated by using Maximal Ratio combining.

- The information from all branch is combined in order to maximize the O/P SNR.



The resulting signal envelop applied to detector

$$r_m = \sum_{i=1}^{n} G_i r$$

Total Noise power

$$N_t = N \sum_{i=1}^{M} G_i^2$$

SNR applied to detector

$$\gamma_m = r_m^2 / 2N_t$$

Equal gain Combining:

- Equal gain Combining is similar to Maximal Ratio Combining without weighting the signals before Summation.
- In EGC Co-phasing is needed to avoid signal Cancellation.
- The average SNR improvement of EGC is typically about 1 dB worse than with MRC but still simpler to implement than MRC.

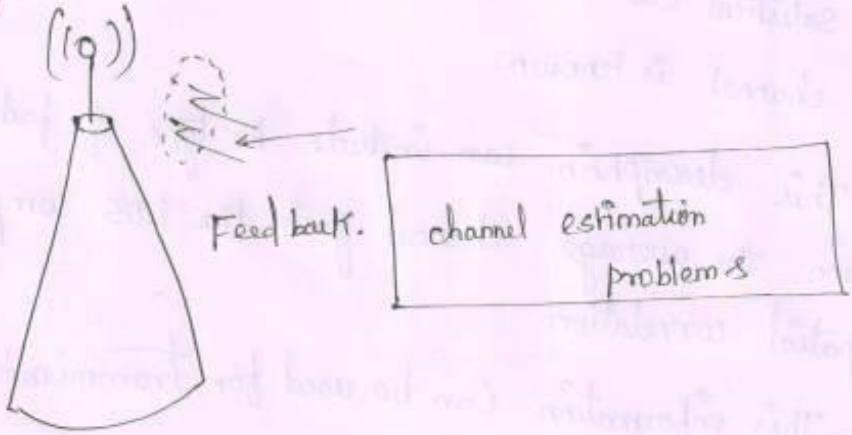
channel state information (11)

- In wireless Communication channel state information simply represents the properties of a Communication link b/w the transmitter and receiver.

- The CSI describes how a signal propagates from the transmitter to the receiver & represents the combined effect of for ex scattering, fading & power delay with distance.

- The CSI makes it possible to adapt transmission to current channel conditions which is crucible for achieving reliable communication with high data rates in multiantenna systems.

- The CSI at the transmitter is vital in MIMO systems in order to increase the transmission rate, to enhance coverage, to improve spectral efficiency and to reduce receiver complexity.



- The CSI is usually estimated at the receiving end & then quantized & fed back to transmitting side.

Instaneous CSI

- It is also known as Short term CSI.
- CSI means that the current conditions of the channel are known which can be viewed as knowing the impulse response of a digital filter.
- This gives an opportunity to adapt the transmitted signal to the impulse response & thereby optimize the received signal for spatial multiplexing or to achieve low bit error rates.

Statistical CSI

- It is known as long-term CSI.
- Satistical CSI means that a statistical characterization of the channel is known.
- This description can include the type of fading distribution the average channel gain the LOS component & the spatial correlation.
- This information can be used for transmission

- The capacity of a MIMO channel is influenced by the degree of CSI available of both transmitter & receiver. (12)

- In most instances of multi antenna communication the receiver can accurately track the instantaneous state of the channel from pilot signals that are typically embedded within the transmission.

Capacity in fading & Non fading channel

MIMO capacity: channel Unknown at the transmitter

- The generalized capacity equation for time space architecture.

- The transmitter only knows the channel satistics such as distributions of the channel distribution parameters.

$$C = \log_2 \left[I_N + \left(\rho / M \right) H H^{(+)} \right] b/s/Hz$$

where $(+)$, H, I_N & ρ represents transpose conjugate, $N \times M$ channel Matrix, $N \times N$ identity matrix & SNR.

The capacity of a MIMO system improves linearly with m fold where $m = \min(M, N)$

MIMO capacity channel known at the transmitter

- The additional performance gain can be achieved in MIMO systems with the CSI at the transmitter
- This scenario considers that the transmitter knows the random channel outcomes & adjust the transmit signal.

$$C = \log_2 [I_N + HQH(\dagger)] \text{ b/s/Hz}$$

If Q denotes the covariance matrix of the transmitted M-D vector Gaussian signal of total radiated power P then the Shannon's capacity for a fading MIMO channel with AWGN is given as.

where $(\dagger)$, H & I_N represents the determinant transpose conjugate, NxM channel matrix & NxN identity matrix

$$Q = (P/M) I_N$$